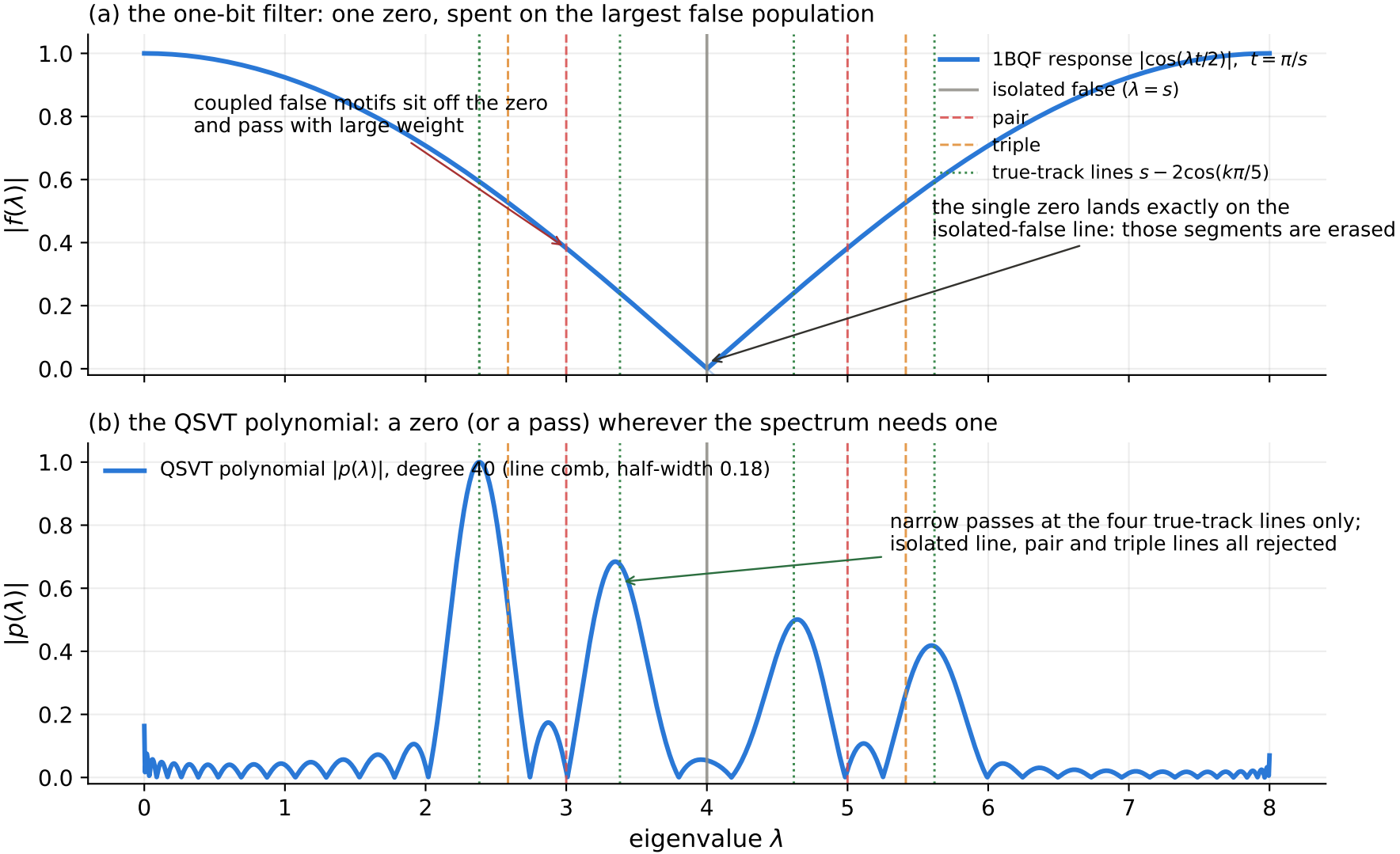
(a) the one-bit filter: one zero, spent on the largest false population
1.0
0.8
0.6
0.4
0.2
0.0
$|f(\lambda)|$
coupled false motifs sit off the zero
and pass with large weight
1BQF response $|\cos(\lambda t/2)|$, $t = \pi/s$
isolated false ($\lambda = s$)
pair
triple
true-track lines $s - 2\cos(k\pi/5)$
the single zero lands exactly on the
isolated-false line: those segments are erased
(b) the QSVT polynomial: a zero (or a pass) wherever the spectrum needs one
QSVT polynomial $|p(\lambda)|$, degree 40 (line comb, half-width 0.18)
narrow passes at the four true-track lines only;
isolated line, pair and triple lines all rejected
$|p(\lambda)|$
0
1
2
3
4
5
6
7
8
eigenvalue λ